

A top-down view of various school supplies arranged on a white surface. In the top left, a black mesh pen holder contains several pens and pencils. Below it, a pair of blue-handled scissors is open. To the right of the scissors is a red pencil, a white eraser, and a small blue sharpener. Further right is a black calculator with a small screen and various function buttons. Next to the calculator is a blue stapler. At the bottom, a long blue ruler with white markings is laid out. Above the ruler, there are two protractors (one blue, one white) and a pink sticky note. The right side of the image features a blue gradient background with text overlays.

Week-8

Introduction to Database Systems

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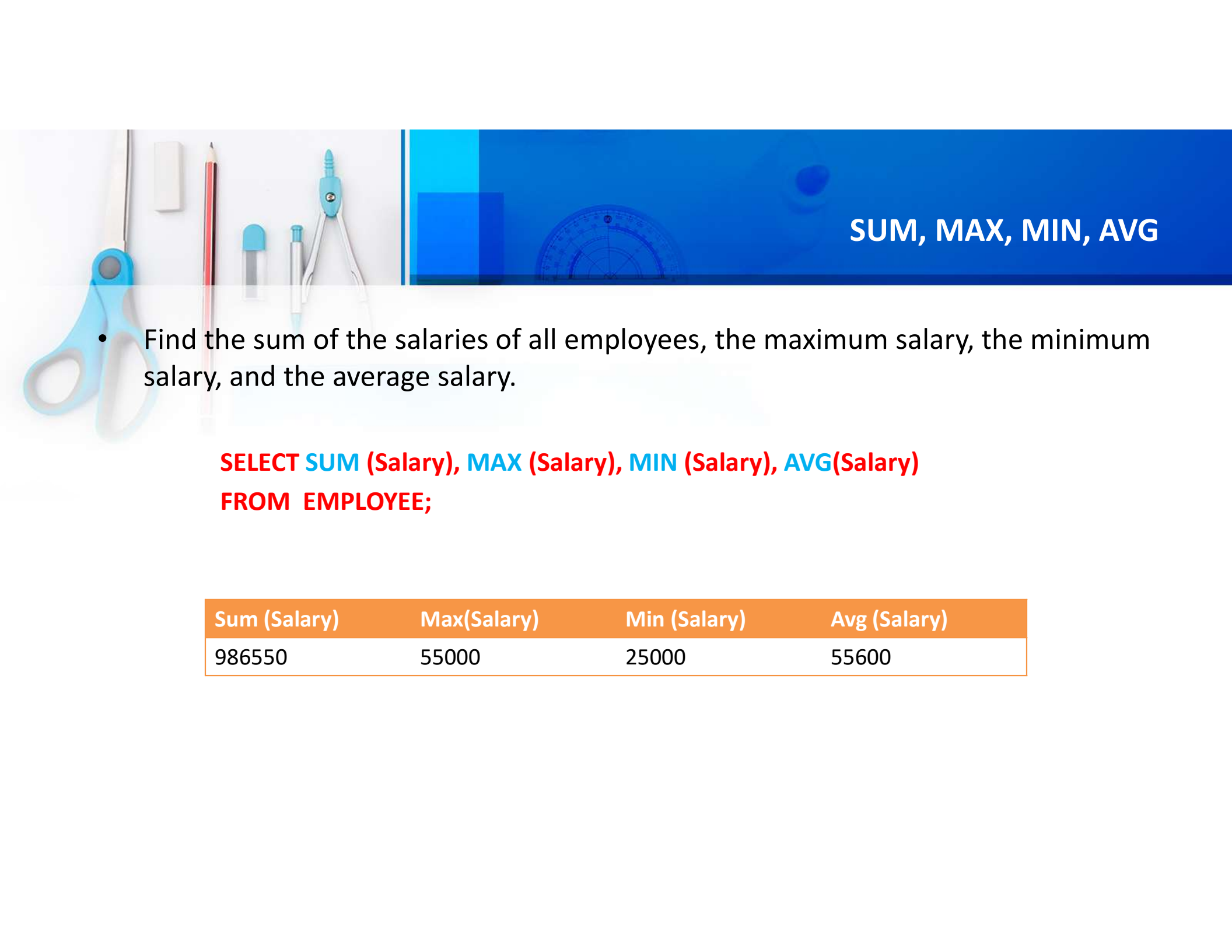
Aggregate Functions
SUM, MIN, MAX
AVG, COUNT, DISTINCT
GROUP BY,

Relational Algebra



Introduction

- An aggregate function performs a calculation on a set of values of a column, and returns a single value
- Built-in aggregate functions
COUNT, SUM, MAX, MIN, and AVG
- Functions can be used in the SELECT clause or in a HAVING clause

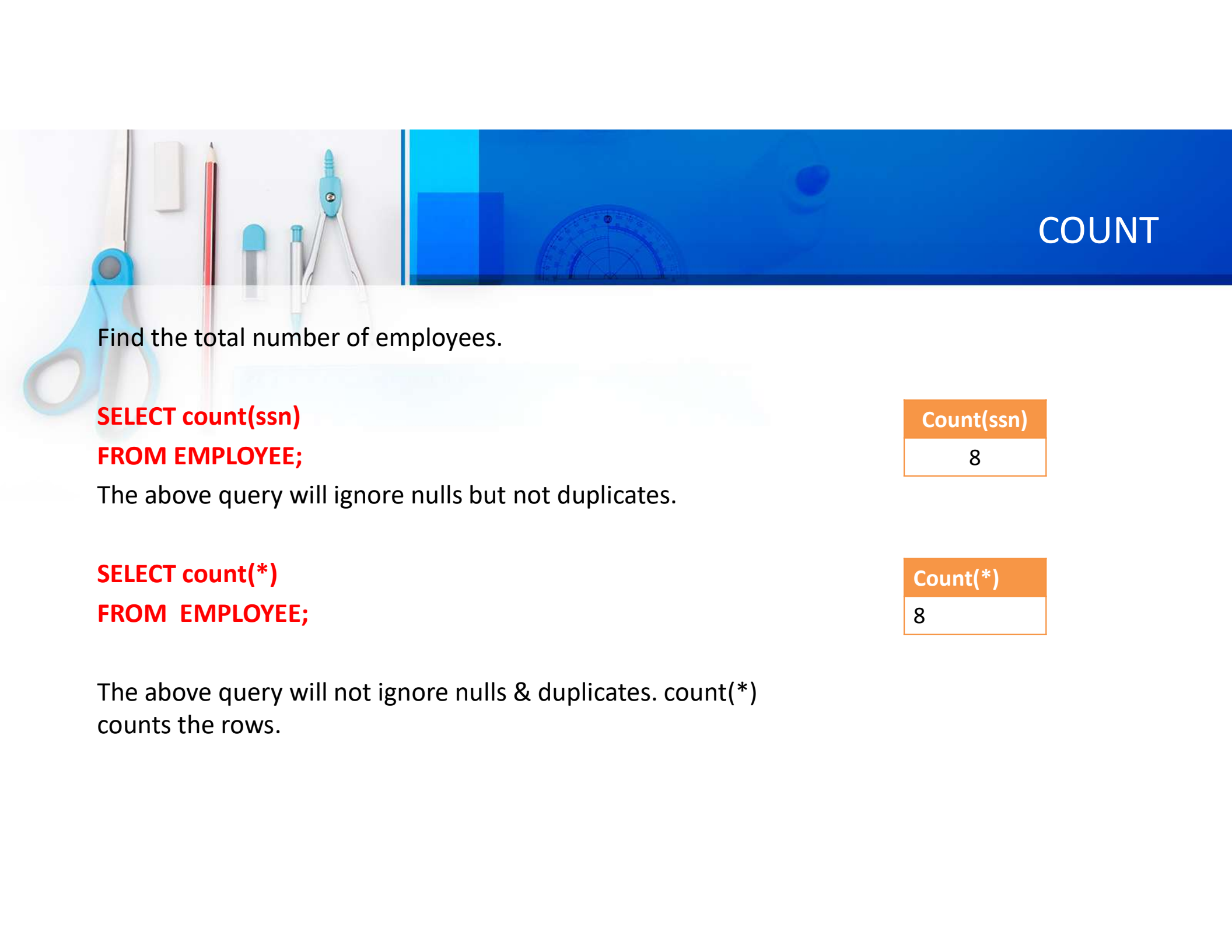


SUM, MAX, MIN, AVG

- Find the sum of the salaries of all employees, the maximum salary, the minimum salary, and the average salary.

```
SELECT SUM (Salary), MAX (Salary), MIN (Salary), AVG(Salary)  
FROM EMPLOYEE;
```

Sum (Salary)	Max(Salary)	Min (Salary)	Avg (Salary)
986550	55000	25000	55600



COUNT

Find the total number of employees.

```
SELECT count(ssn)  
FROM EMPLOYEE;
```

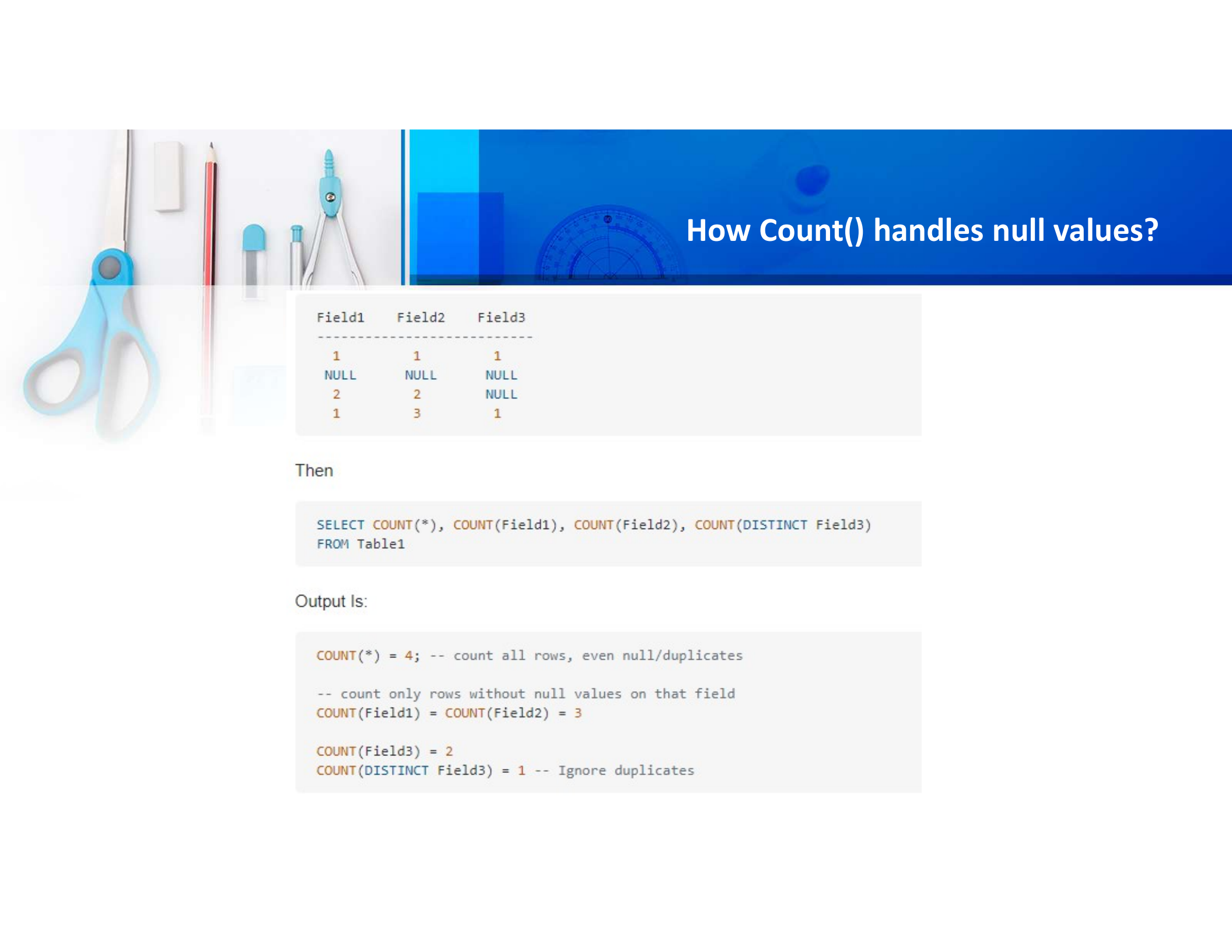
The above query will ignore nulls but not duplicates.

```
SELECT count(*)  
FROM EMPLOYEE;
```

The above query will not ignore nulls & duplicates. count(*) counts the rows.

Count(ssn)
8

Count(*)
8



How Count() handles null values?

Field1	Field2	Field3
1	1	1
NULL	NULL	NULL
2	2	NULL
1	3	1

Then

```
SELECT COUNT(*), COUNT(Field1), COUNT(Field2), COUNT(DISTINCT Field3)
FROM Table1
```

Output Is:

```
COUNT(*) = 4; -- count all rows, even null/duplicates

-- count only rows without null values on that field
COUNT(Field1) = COUNT(Field2) = 3

COUNT(Field3) = 2
COUNT(DISTINCT Field3) = 1 -- Ignore duplicates
```

DISTINCT

Count the number of distinct salary values in the database

```
SELECT    COUNT (DISTINCT CUST_CODE) AS  
"Number of Employees"  
FROM      ORDERS;
```

ORD_NUM	AGENT_CODE	CUST_CODE
200101	A008	C00001
200114	A008	C00002
200122	A004	C00003
200134	A005	C00004
200106	A002	C00005
200104	A004	C00006
200135	A010	C00007
200119	A010	C00007
200107	A010	C00007
200108	A004	C00008
200121	A004	C00008
200120	A002	C00009
200133	A002	C00009
200128	A002	C00009
200116	A009	C00010
200109	A010	C00011
200102	A012	C00012
200131	A012	C00012

** for better understand
the table have been arranged on
cust_code



Number of employees
25

result



count once

Group By

- Aggregate functions are often used with the GROUP BY clause
- GROUP BY Clause is used to collect data from multiple records and group the result by one or more column
- **Example:** "find the number of authors in each country".

```
SELECT column_name(s)
FROM table_name
WHERE condition
GROUP BY column_name(s)
ORDER BY column_name(s);
```

```
Select Country, count(*)
From Author
Group by country;
```

Here is the Output

aut_id	country
AUT011	Australia
AUT013	Australia
AUT009	Brazil
AUT002	Canada
AUT012	Canada
AUT005	Germany
AUT004	India
AUT007	UK
AUT014	UK
AUT003	UK
AUT001	UK
AUT010	USA
AUT008	USA
AUT006	USA
AUT015	USA

country	COUNT(*)
Australia	2
Brazil	1
Canada	2
Germany	1
India	1
UK	4
USA	4

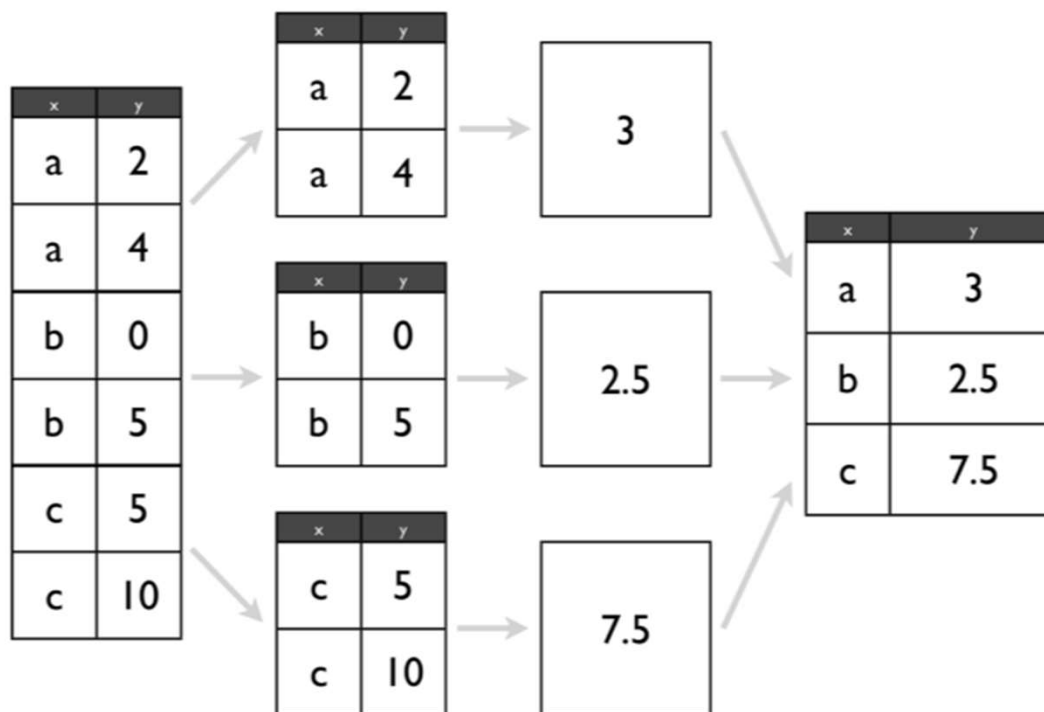
group of country

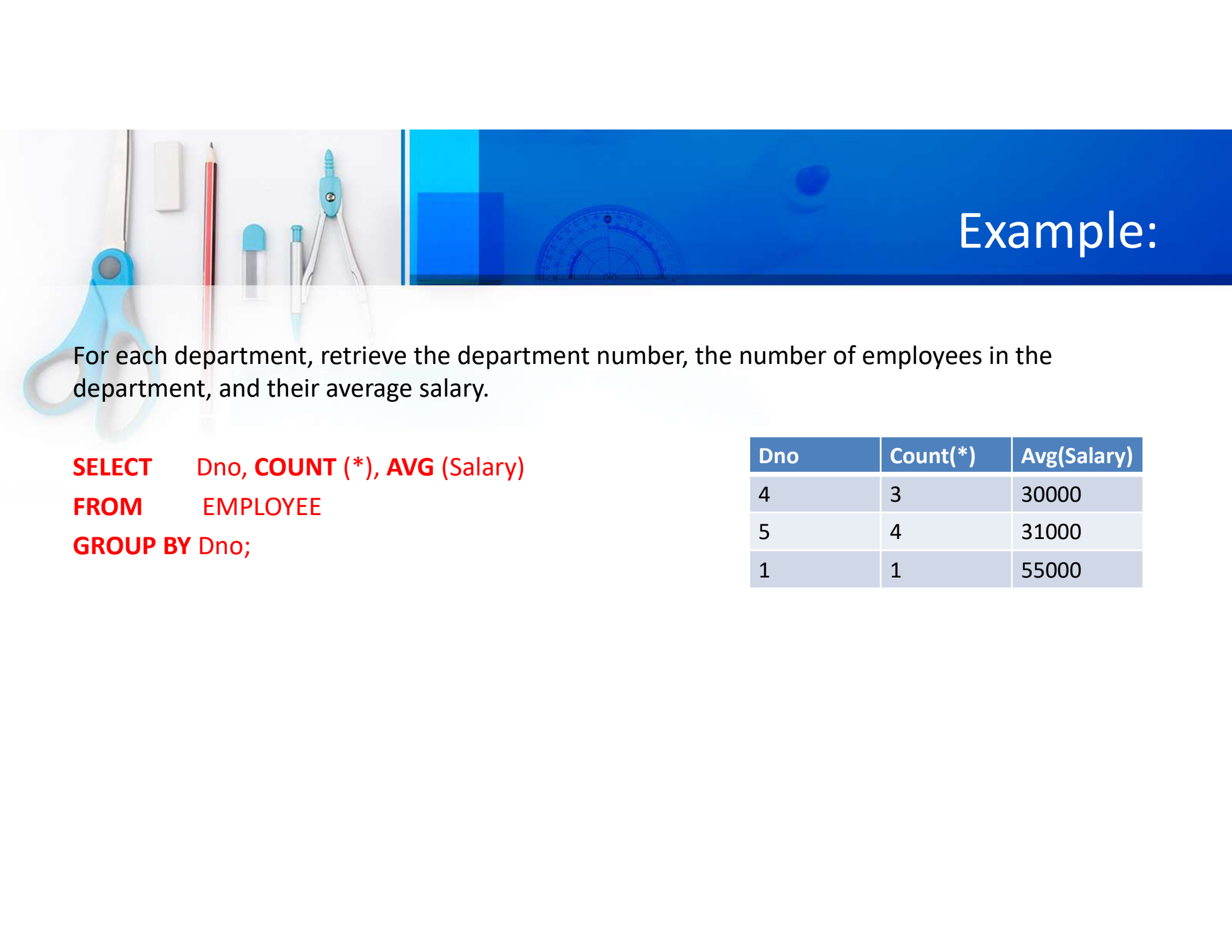
Group By

Split

Apply

Combine



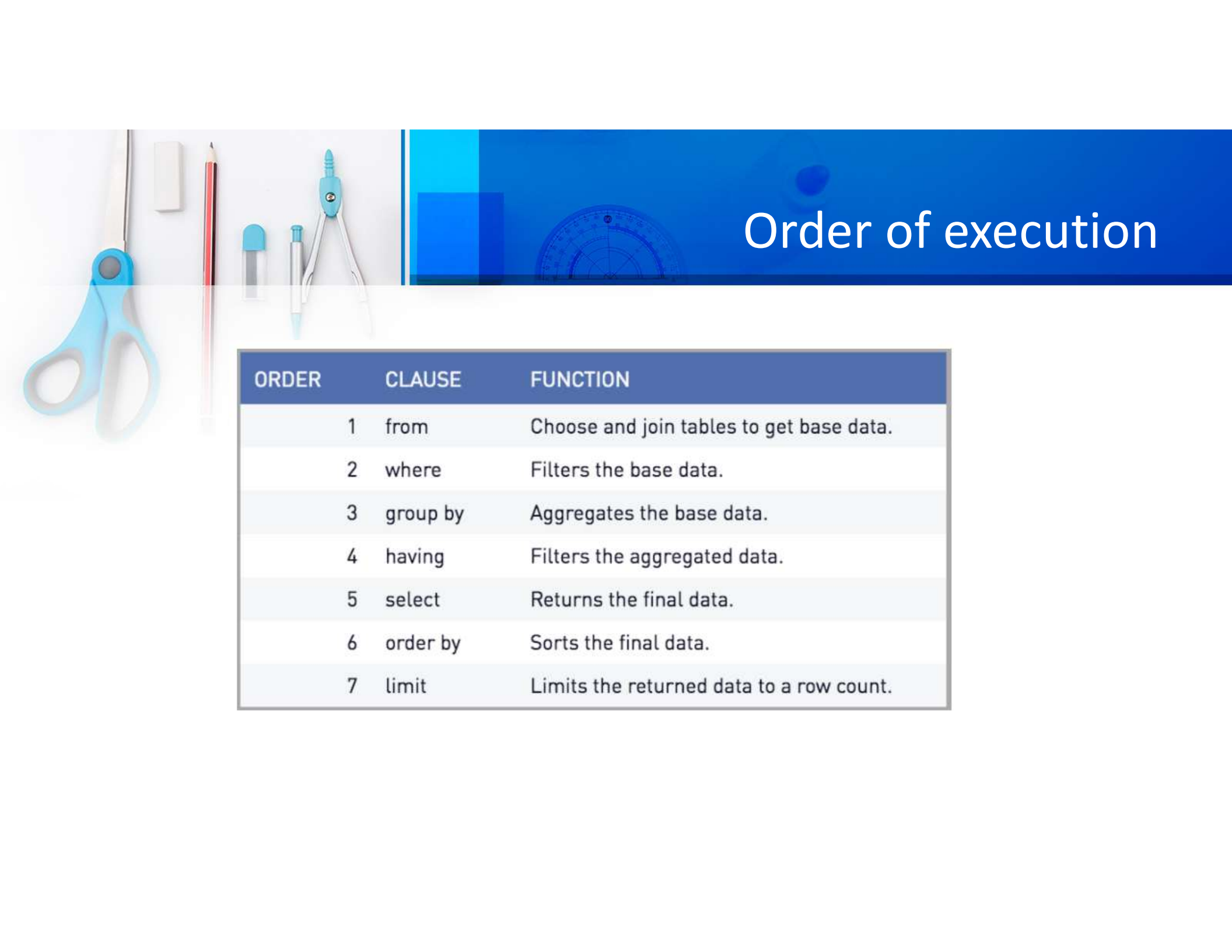


Example:

For each department, retrieve the department number, the number of employees in the department, and their average salary.

```
SELECT    Dno, COUNT (*), AVG (Salary)
FROM      EMPLOYEE
GROUP BY Dno;
```

Dno	Count(*)	Avg(Salary)
4	3	30000
5	4	31000
1	1	55000



Order of execution

ORDER	CLAUSE	FUNCTION
1	from	Choose and join tables to get base data.
2	where	Filters the base data.
3	group by	Aggregates the base data.
4	having	Filters the aggregated data.
5	select	Returns the final data.
6	order by	Sorts the final data.
7	limit	Limits the returned data to a row count.



Relational Algebra



Sum

Example:

```
SELECT SUM (Salary) AS TotalSalary FROM EMPLOYEE;
```

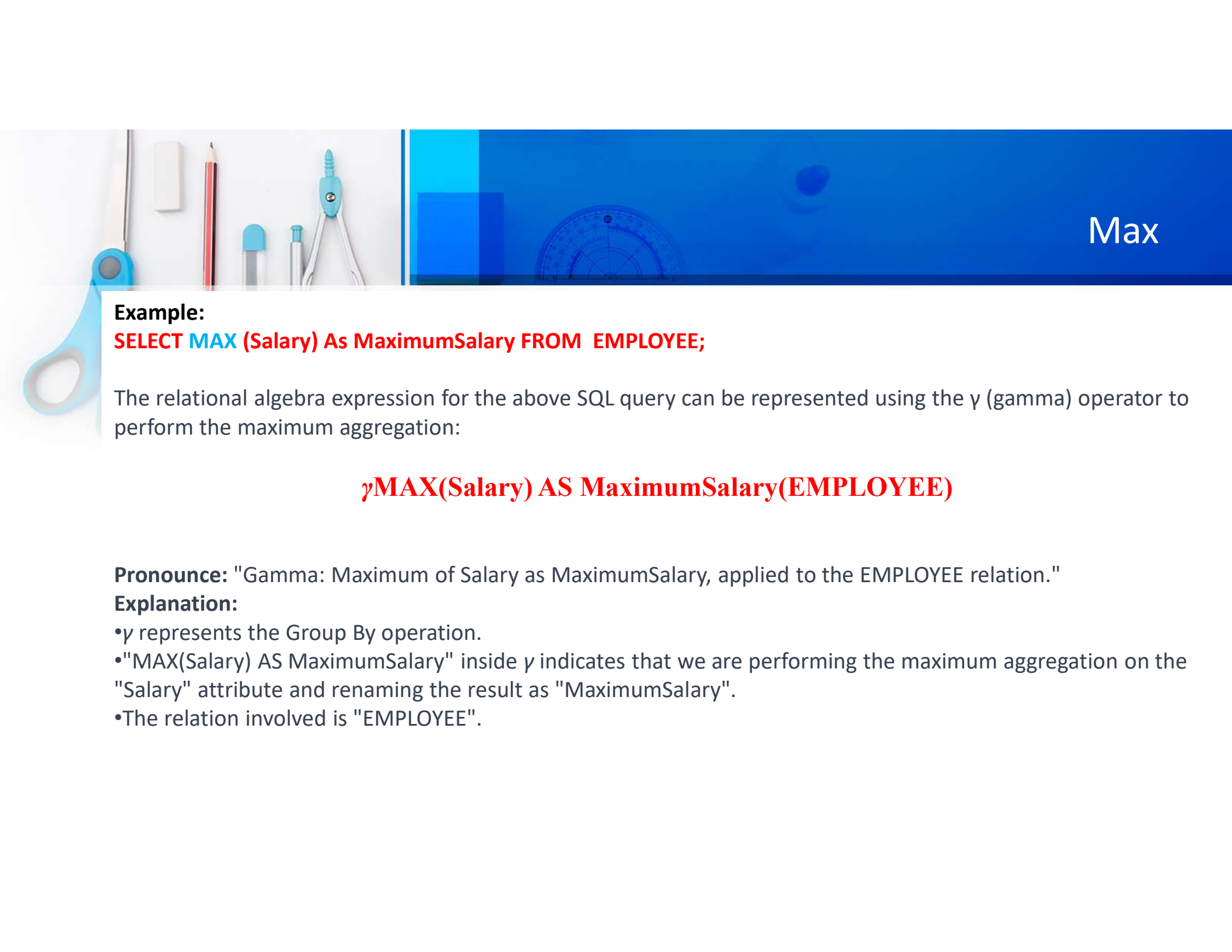
In relational algebra, the expression for the above SQL query can be represented using the γ (gamma) operator to perform the summation:

γ SUM(Salary) AS TotalSalary(EMPLOYEE)

Pronunciation: "Gamma: Sum of Salary as TotalSalary, applied to the EMPLOYEE relation."

Explanation:

- γ represents the Group By operation.
- "SUM(Salary) AS TotalSalary" inside γ indicates that we are performing the sum aggregation on the "Salary" attribute and renaming the result as "TotalSalary".
- The relation involved is "EMPLOYEE".



Max

Example:

SELECT MAX (Salary) AS MaximumSalary FROM EMPLOYEE;

The relational algebra expression for the above SQL query can be represented using the γ (gamma) operator to perform the maximum aggregation:

γ MAX(Salary) AS MaximumSalary(EMPLOYEE)

Pronounce: "Gamma: Maximum of Salary as MaximumSalary, applied to the EMPLOYEE relation."

Explanation:

- γ represents the Group By operation.
- "MAX(Salary) AS MaximumSalary" inside γ indicates that we are performing the maximum aggregation on the "Salary" attribute and renaming the result as "MaximumSalary".
- The relation involved is "EMPLOYEE".



Min

Example:

SELECT MIN (Salary) AS MinimumSalary FROM EMPLOYEE;

The relational algebra expression for the above SQL query can be represented using the γ (gamma) operator to perform the maximum aggregation:

γ MIN(Salary) AS MinimumSalary(EMPLOYEE)

Pronounce: "Gamma: Minimum of Salary as MinimumSalary, applied to the EMPLOYEE relation."

Explanation:

- γ represents the Group By operation.
- "MIN(Salary) AS MinimumSalary" inside γ indicates that we are performing the minimum aggregation on the "Salary" attribute and renaming the result as "MinimumSalary".
- The relation involved is "EMPLOYEE".

Example:

```
SELECT AVG (Salary) AS AverageSalary FROM EMPLOYEE;
```

The relational algebra expression for the above SQL query can be represented using the γ (gamma) operator to perform the maximum aggregation:

γ AVG(Salary) AS AverageSalary(EMPLOYEE)

Pronounce: "Gamma: Average of Salary as AverageSalary, applied to the EMPLOYEE relation."

Explanation:

- γ represents the Group By operation.
- "MIN(Salary) AS AverageSalary" inside γ indicates that we are performing the Average aggregation on the "Salary" attribute and renaming the result as "AverageSalary".
- The relation involved is "EMPLOYEE".

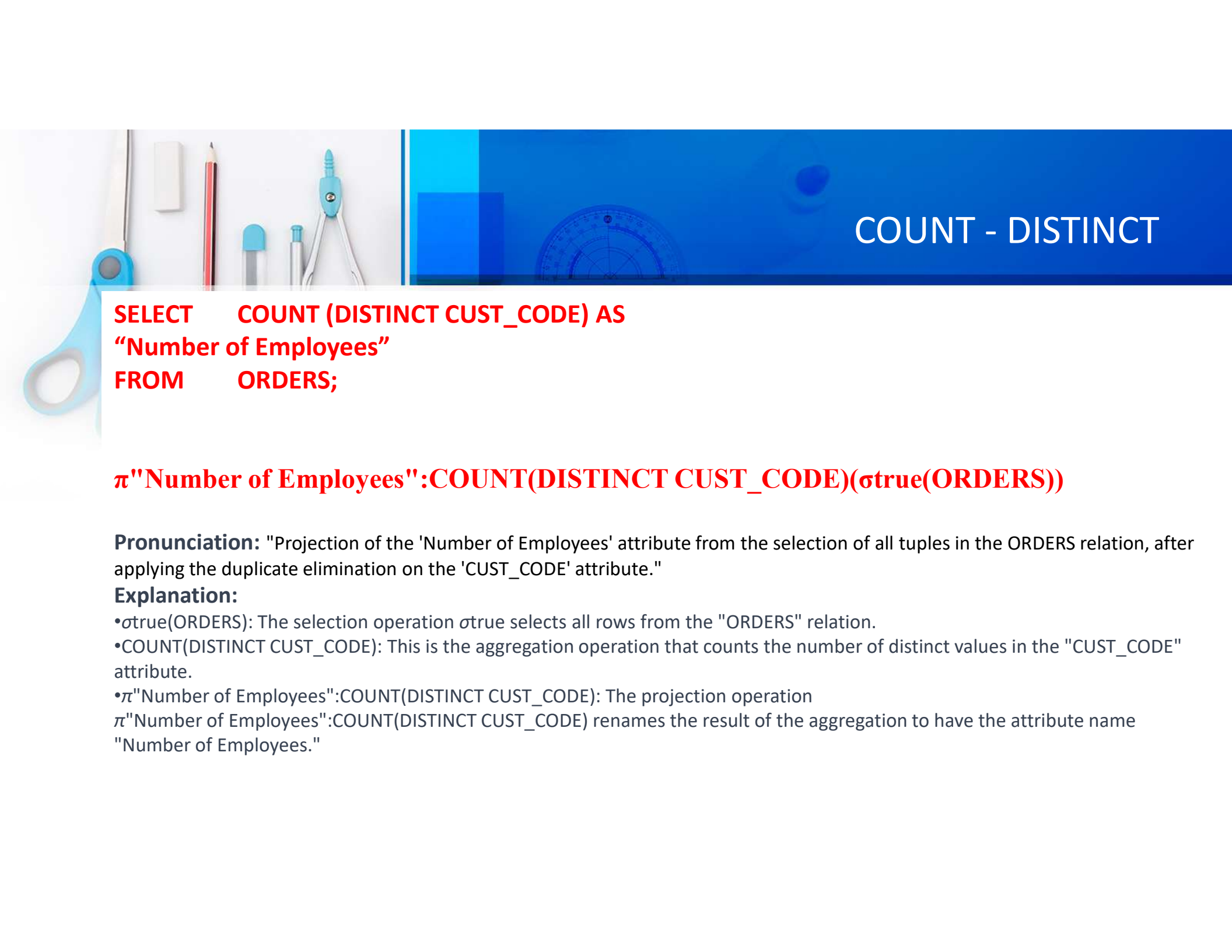
```
SELECT    COUNT (CUST_CODE) AS  
"Number of Employees"  
FROM      ORDERS;
```

π "Number of Employees"(δ CUST_CODE(σ true(ORDERS)))

Pronunciation: "Projection of the 'Number of Employees' attribute from the selection of all tuples in the ORDERS relation, after applying the duplicate elimination on the 'CUST_CODE' attribute."

Explanation:

- σ true(ORDERS): This is the selection operation with a condition that is always true, effectively representing the entire ORDERS relation.
- δ CUST_CODE(...): This is the duplicate elimination operation, ensuring that only distinct values of CUST_CODE are considered.
- π "Number of Customers"(...): This is the projection operation that selects only the count of distinct CUST_CODE values, renaming it as "Number of Customers."



COUNT - DISTINCT

```
SELECT    COUNT (DISTINCT CUST_CODE) AS  
"Number of Employees"  
FROM      ORDERS;
```

π "Number of Employees":COUNT(DISTINCT CUST_CODE)(σ_{true} (ORDERS))

Pronunciation: "Projection of the 'Number of Employees' attribute from the selection of all tuples in the ORDERS relation, after applying the duplicate elimination on the 'CUST_CODE' attribute."

Explanation:

- σ_{true} (ORDERS): The selection operation σ_{true} selects all rows from the "ORDERS" relation.
- COUNT(DISTINCT CUST_CODE): This is the aggregation operation that counts the number of distinct values in the "CUST_CODE" attribute.
- π "Number of Employees":COUNT(DISTINCT CUST_CODE): The projection operation π "Number of Employees":COUNT(DISTINCT CUST_CODE) renames the result of the aggregation to have the attribute name "Number of Employees."



Group By

Example:

```
SELECT CustomerID, COUNT(*) AS OrderCount  
FROM Orders GROUP BY CustomerID;
```

```
 $\pi$ CustomerID, OrderCount( $\gamma$ CustomerID; COUNT(*) AS OrderCount(Orders))
```

Pronunciation: "Projection of 'CustomerID' and 'OrderCount' from the result of grouping Orders by 'CustomerID' and applying the COUNT aggregate function, with the result aliased as 'OrderCount'."

Explanation:

- γ CustomerID; COUNT(*) AS OrderCount(Orders): This is the grouping operation where we group the rows of the Orders table by CustomerID and calculate the count of rows (orders) for each group.
- π CustomerID, OrderCount(...): This is the projection operation that selects only the CustomerID and OrderCount columns from the result of the grouping operation.



Practice

Group By Subject count ?

Group By Year Count?

SUBJECT	YEAR	NAME
English	1	Harsh
English	1	Pratik
English	1	Ramesh
English	2	Ashish
English	2	Suresh
Mathematics	1	Deepak
Mathematics	1	Sayan

The header features a horizontal strip. On the left, there is a photograph of school supplies: a pair of blue-handled scissors, a white eraser, a red pencil, a blue sharpener, and a pair of compasses. To the right of this is a solid blue background with a faint, light blue geometric pattern consisting of squares and a circular arc.

Practice:

- For each project, retrieve the project number, the project name, and the number of employees who work on that project.

```
SELECT
    p.project_number,
    p.project_name,
    COUNT(e.employee_id) AS num_employees
FROM
    projects p
JOIN
    employees e ON p.project_number = e.project_number
GROUP BY
    p.project_number, p.project_name;
```



Practice:

- For each project, retrieve the project number, the project name, and the number of employees who work on that project.



- For each project, retrieve the project number, the project name, and the number of employees from department 5 who work on the project.



- For each project, retrieve the project number, the project name, and the number of employees from department 5 who work on the project.

Select pnumber, pname, count(essn)

From (project join workson on pno= pnumber) join employee on essn =ssn

Where dno = 5

group by pnumber;

The header features a white background on the left with various school supplies: a pair of blue-handled scissors, a white eraser, a red pencil, a blue sharpener, a blue pen, and a pair of compasses. To the right, a blue gradient background contains a faint, light blue geometric diagram of a circle with internal lines and a semi-circular arc. The title 'Practice Queries' is written in white serif font on the right side of the blue background.

Practice Queries

1. Show average salary of all employees.
2. Show total number of departments.
3. Show sum of hours spent on all projects.
4. Show average salary of each department if average salary is greater than 40,000.
5. Show minimum salary of each department if minimum salary is less than 20,000.
6. Show total number of employees of each department with department name.
7. Show number of dependent of each employee.
8. Show count of projects of all departments in ascending order by department name.
9. Show count of total locations of each department which are located in Lahore.
10. For each project show project name and total number of employees working on it.



Thank You all!